The Impact and Future Adoption of Vision-Based Hand Gesture Recognition in Touch-Free Human-Computer Interaction: A Survey-Based Study

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***Abstract*—Vision-based hand gesture recognition is moving fast. Thanks to deep learning models like YOLO and convolutional classifiers, standard consumer webcams can now track movements in real time. But look at most published papers in this field. They almost always measure success through raw detection accuracy, frame rates, and how well the system holds up under tightly controlled lab conditions. This leaves a massive question completely unexamined: do regular people actually trust, understand, or even want to use these systems?**

We wanted to find out. This paper breaks down a survey-based study exploring how users perceive vision-based gesture recognition as a touchless alternative to the classic mouse and keyboard setup. We gathered eighteen participants from a university community in Sri Lanka. They filled out a structured questionnaire covering their daily computer habits, physical discomfort from current devices, and whether they had ever even encountered gesture controls before. We also looked at perceived trust, utility, preferred functions, and the barriers keeping them from adopting the tech.

The results were eye-opening. While the participants aren't very familiar with gesture control, they are incredibly open to it. In fact, two-thirds of the respondents admit to feeling physical discomfort from traditional keyboards and mice. More than half said they would happily switch to gesture control if it were available right now. Not a single person said they would reject it completely.

Right now, the real hesitation isn't about attitude—it's purely technical. Users care way more about precision, response speed, and data privacy than how hard the system is to learn. When we look at these findings alongside traditional engineering literature, the takeaway is clear. Closing the technical accuracy gap in the lab is the single most direct way to close the adoption gap in the real world.

Keywords—computer interaction; hand gesture recognition; touch-free interaction; technology acceptance; user perception; survey research; computer vision

I. INTRODUCTION (*HEADING 1*)

For nearly half a century, the mouse and keyboard have defined how human beings interact with computers. They get the job done. Yet, neither device was actually designed to withstand the physical toll of modern eight-hour workdays or nonstop screen time. Frequent computer users routinely face repetitive wrist strain, finger fatigue, eye strain, and poor posture from keeping their hands locked in the same position for hours on end. The group surveyed for this study is no exception. In fact, as later sections reveal, two out of three respondents already suffer from some form of physical discomfort tied directly to their current input devices.

Touchless interaction has emerged as a promising way to break this cycle, bringing options like voice assistants, eye-gaze tracking, and vision-based gesture recognition to the table. Gesture control in particular has captured the research community's attention because hands are our natural, built-in communication tools; humans wave, point, and shape their fingers to express meaning long before anyone teaches them how to type. Recent breakthroughs in object detection and convolutional neural networks (CNNs) have finally pushed these vision-based systems out of the realm of lab curiosities. They now run smoothly on standard laptop webcams. A perfect example is the YOLO-CNN pipeline analyzed in Section II, which achieves real-time tracking at 25 to 48 frames per second alongside a top-1 classification accuracy exceeding 93%—all using a basic, off-the-shelf RGB camera.

This technical progress is massive, but it only answers half the question. Real-world adoption demands more than just laboratory success. An algorithm can be blindingly fast and incredibly precise, but it will still fail if the target users do not trust it, do not know it exists, or simply cannot see how it fits into their daily routines. History proves this point. Even established technologies like touchscreens and voice assistants went through long awkward phases where the public remained deeply unfamiliar with them, despite the underlying engineering being fully mature. Whether gesture-based interaction faces that same hurdle—and what specifically triggers user hesitation—is something raw accuracy metrics cannot uncover.

Right now, the vast majority of hand gesture research is fixated on algorithmic optimization. Engineers spend their time boosting classification rates and building system resilience against challenging environmental conditions. Far fewer studies pause to ask how everyday, non-expert users actually perceive, trust, and plan to use these systems once they leave the laboratory. Granted, there are isolated pieces of literature exploring the acceptability of mid-air or around-device gestures, alongside trust metrics for gesture controls in high-stakes environments like driving. But these studies remain rare. They seldom address the core question driving this paper: how do everyday computer users—not tech experts or lab subjects—feel about using vision-based hand gestures to completely replace the mouse and keyboard they rely on every single day? This study addresses that exact knowledge gap using fresh survey data from a university community in Sri Lanka.

To be perfectly clear, the contribution of this paper is not a new, working gesture recognition system. We did not build software, write code, or gather machine-learning data for this project; all performance figures cited here belong to the existing research literature and serve strictly as a benchmark for comparison. Instead, this paper contributes a detailed, data-driven narrative from a specific demographic of computer users. We lay out exactly what they already know, what it would take to earn their trust, the specific barriers holding them back, and the exact tasks they want a gesture system to handle.

The layout of the paper follows a straightforward path. Section II reviews the perception-focused literature to anchor our study in current research. Section III breaks down our survey instrument, our sampling choices, and how we analyzed the data. Moving on, Section IV presents our core survey findings with detailed commentary, while Section V maps out real-world application areas suggested by both past literature and our respondents. Section VI tackles the engineering and social challenges blocking everyday adoption. Finally, Sections VII and VIII outline our study's limitations and pitch future research directions, leading into our closing thoughts in Section IX.

II. LITERATURE REVIEW

A. *Technical Advances in Vision-Based Gesture Recognition*

Early gesture recognition relied heavily on hand-crafted features. For instance, Zhang and Lu utilized color-histogram segmentation paired with template matching [25]. This approach yielded decent results under controlled lighting conditions, but performance plummeted the moment the background color closely mirrored the user's skin tone. Starner and Pentland shifted the paradigm by introducing Hidden Markov Models for American Sign Language recognition, establishing a probabilistic framework that heavily influenced subsequent research [22]. Later on, the debut of depth sensors like the Microsoft Kinect enabled Khan and Kim to merge depth and RGB data through convolutional networks [13]. This combination made systems far more resilient to varied hand orientations, though it introduced a frustrating dependency on specialized, expensive hardware.

The shift toward deep learning changed everything. Fang et al. demonstrated that a lightweight CNN could comfortably outperform SVM-based classifiers while remaining fast enough for real-time operations [9]. Subsequent iterations integrated attention mechanisms and residual connections to catch micro-movements in finger joints. For dynamic, motion-heavy gestures, LSTM networks coupled with CNN feature extractors allowed researchers to model temporal sequences directly [12]. Jang and Choi pushed this further into a spatiotemporal framework, validating it on the rigorous ChaLearn dataset [24].

Yet, an underlying flaw persisted. In their survey of deep-learning-based gesture recognition, Park and Han pointed out that many systems naively assume the hand is already perfectly cropped and isolated from the scene before

classification even begins [18]. It is an assumption that rarely holds up in messy, real-world video feeds. The rollout of YOLO by Redmon et al. tackled this bottleneck head-on by reframing detection as a pure regression problem, cracking speeds over 45 FPS on standard GPU hardware [21]. Hybrid YOLO-CNN pipelines built on this breakthrough—including the baseline architecture informing this study—now clock accuracy rates north of 93% at 25 to 48 FPS [10]. Best of all, they achieve this on standard RGB webcams without needing a single depth sensor.
Pieced together, the engineering literature tells a clear story: accuracy and speed have hit a point where hardware limitations are no longer the bottleneck. What the literature completely glosses over is the human element. We still do not know if the average person sitting in front of a monitor actually wants to wave at it, or if they trust a camera enough to open a file for them.

B. *User Perception and Social Acceptability of Gesture Interaction*

A completely separate branch of HCI research asks a different question. It doesn't look at whether gesture recognition works, but whether people actually want it. Ahlström, Hasan, and Irani investigated the social acceptability of gestures performed around devices in public spaces [1]. Their findings revealed that user comfort hinges heavily on how conspicuous the movement looks to bystanders, completely independent of how well the system registers the command. Koelle, Ananthanarayan, and Boll audited the methodologies used to study acceptability in wearable and gesture-based HCI [15]. They concluded that true adoption depends on social context, public exposure, and perceived system capability, rather than technical maturity alone. Context matters immensely. In an automotive study, Hafizi et al. examined mid-air gestures for in-car interfaces and found they caused significantly more driver distraction than traditional touch inputs [11]—a stark reminder that what works perfectly in an office might fail miserably on the road.
Niche application studies mirror this cautious reality. When Babu, Rustamov, and Turaev pitched a touchless gesture system for healthcare settings, they discovered its real selling point was hygiene and zero-contact safety, not the novelty of the tech itself [2]. Similarly, Brunetti, Gena, and Vernero evaluated interactive tools in human-centered manufacturing environments [4]. They noted that factory workers will only accept a gesture interface if they are entirely confident it won't misfire during high-stakes, time-sensitive tasks. Across all this research, a pattern emerges. Privacy, social appropriateness, and operational reliability consistently outrank raw algorithmic accuracy when users decide whether to adopt a gesture interface. It is a reality that our own survey data explicitly confirms in Section IV.

C. *Theoretical Basis: Technology Acceptance*

The Technology Acceptance Model (TAM) [7] and its evolution, the Unified Theory of Acceptance and Use of Technology (UTAUT) [23], give us a solid framework to make sense of these behavioral patterns. Both models argue that a user's intent to adopt new tech is driven by two main pillars: perceived usefulness and perceived ease of use. These pillars are constantly shaped by trust, risk factors, and, in the case of UTAUT, social influence and facilitating conditions.
These core theoretical constructs map directly onto our survey design. For example, our questions regarding whether gesture controls could realistically replace a mouse test perceived usefulness. Queries drilling down into system trust and accuracy address perceived risk. Meanwhile, tracking where participants first heard about gesture technology allows us to gauge the weight of social influence within their community.

D. *Summary and Research Gap*

Reviewing these two research tracks reveals a deep disconnect rather than an outright contradiction. On one side, the computer vision and engineering communities have cooked up incredibly fast, highly accurate classification models that run flawlessly on everyday consumer hardware, with the YOLO-CNN pipeline serving as a prime example [10], [21]. On the other side, user perception studies remain sparse. They are largely restricted to hyper-specific environments like public kiosks, smart factories, or car dashboards.
Hardly anyone is asking the obvious question: how do regular computer users—not gesture researchers, factory workers, or drivers—view gesture control as a direct, day-to-day replacement for the mouse and keyboard sitting on their desks? This study aims to bridge that exact divide. By surveying a diverse community of computer users at a university in Sri Lanka, we contrast real user feedback directly against the engineering milestones mentioned above, forcing these two isolated conversations into the same room.

III. Methodology

A. *Research Design*

This study employs a cross-sectional survey design. It leans heavily on quantitative data collection but carves out room for qualitative insights through strategically placed open-ended questions. Structurally, the instrument mirrors classic technology acceptance frameworks outlined in Section II. We used structured scalar questions to measure core variables like perceived usefulness, ease of use, trust, and behavioral intent. Meanwhile, the free-text queries allowed participants to articulate specific feature requests and lingering anxieties in their own words.

B. Instrument

Data collection was handled entirely through a structured Google Forms questionnaire broken into five distinct modules:

- **Module I:** Professional and demographic background.
- **Module II:** Current computer usage profiles, primary input configurations, daily exposure hours, and physical pain metrics.
- **Module III:** Prior technological awareness and hands-on exposure to gesture-based interfaces.
- **Module IV:** Psychometric evaluations utilizing a five-point Likert scale (ranging from strongly disagree to strongly agree) to probe trust, utility, perceived novelty, accessibility, and hardware readiness.
- **Module V:** Feature preferences, adoption hurdles, and three qualitative prompts exploring improvement areas, real-world applications, and the long-term trajectory of human-computer interaction.

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C. Participants and Sampling

We distributed the questionnaire within the LNBTI Japanese University community and their immediate social networks in Sri Lanka. Ultimately, eighteen responses were collected (n = 18). The resulting sample consists mostly of students (72.2%) skewing toward the 18–24 age bracket (50.0%) or individuals under 18 (38.9%). This distribution reflects a youthful, tech-adjacent university cohort alongside pre-university peers. Intriguingly, 100% of the respondents operate exclusively within the Windows ecosystem. Rather than reflecting global market shares, this uniform operating system profile represents a distinct sampling constraint. We address this directly in Section VII.

D. Data Analysis

Quantitative responses were processed using descriptive statistical methods. For the Likert-scale items, we avoided the common pitfall of collapsing data into binary "agree/disagree" buckets; preserving the full five-point spectrum was essential for capturing subtle shifts in user sentiment. Conversely, the qualitative text fields underwent a systematic thematic analysis. We read through the raw open-ended responses multiple times, assigned initial codes, and iteratively grouped them into broader thematic categories. This workflow yielded six core themes, which we break down in Section IV-I: accuracy, privacy, voice integration, accessibility, lighting dependencies, and natural usability.

E. Ethical Considerations

Participation was entirely voluntary. To protect respondent anonymity, the survey architecture was built without tracking personal identifiers like names or email addresses. Furthermore, every participant reviewed a digital informed consent brief explaining the study's scope and data handling procedures before gaining access to the questionnaire.

IV. Results and Discussion

A. Demographic and Usage Profile

Table I lays out the complete age and occupational breakdown of the survey cohort. The data confirms a student-dominated sample (72.2%), with age concentrations split between the 18–24 bracket (50.0%) and under-18 users (38.9%). Female respondents made up the clear majority at 72.2%. As noted previously, every single participant reported using Windows as their primary operating system. This structural homogeneity limits the immediate generalizability of our findings to alternative ecosystems, a point we revisit as a limitation in Section VII.

Table1 **PARTICIPANT DEMOGRAPHICS**

AGE GROUP	%	OCCUPATION	%
UNDER 18	38.9	STUDENT	72.2
18–24	50.0	TEACHER / LECTURER	5.6
25–34	5.6	OTHER	11.1
Gender: Male 27.8% / Female 72.2%		Operating System: 100% Windows	

<DIV STYLE="TEXT-ALIGN: CENTER;"> <P><B>FIG. 1.</B> AGE DISTRIBUTION OF SURVEY PARTICIPANTS (N = 18).</P> <IMG SRC="MEDIA/IMAGE1.PNG" ALT="AGE DISTRIBUTION PIE CHART" STYLE="WIDTH: 4.58IN; HEIGHT: 2.65IN;"> </DIV>

DATA IN TABLE II HIGHLIGHTS INTENSIVE DAILY COMPUTER USE AMONG A MAJORITY OF USERS, WITH OVER TWO-THIRDS EXCEEDING TWO HOURS OF DAILY SCREEN TIME. CONSIDERING THIS HEAVY USAGE, THE HIGH REPORTED PREVALENCE OF 66.7% FOR PHYSICAL DISCOMFORT RELATED TO INPUT DEVICES IS STRIKING, CHARACTERIZED PRIMARILY BY EYE STRAIN (44.4%), FOLLOWED CLOSELY BY WRIST AND FINGER PAIN (27.8% EACH) AND NECK/SHOULDER FATIGUE (22.2% EACH).

TABLE 2 **COMPUTER USAGE PROFILE**

Usage Metric	%	Notes
Daily computer use	55.6	vs. 33.3% several times/week, 11.1% rarely
Main input device: Mouse	38.9	Touchpad 33.3%, keyboard shortcuts 27.8%

2–4 hrs/day on computer	38.9	4–6 hrs 27.8%, <2 hrs 22.2%, >6 hrs 11.1%
Discomfort with input devices	66.7	Eye strain most cited (44.4%)

THE CONCURRENT PRESENCE OF EYE STRAIN AND MUSCULOSKELETAL ISSUES INDICATES THAT PHYSICAL DISCOMFORT STEMS FROM MULTIPLE FACTORS; SCREEN FATIGUE IS A MAJOR COMPONENT THAT GESTURE RECOGNITION SIMPLY CANNOT SOLVE ON ITS OWN. STILL, THE TWO MOST PERVASIVE ISSUES—WRIST STRAIN AND FINGER PAIN—FLOW DIRECTLY FROM THE REPETITIVE MECHANICS OF USING A TRADITIONAL MOUSE AND KEYBOARD. THIS IS EXACTLY WHERE TOUCHLESS SYSTEMS CAN STEP IN TO ALTER USER ERGONOMICS. THESE FINDINGS ALIGN WITH BROADER HCI LITERATURE ON CAMERA-BASED INTERFACES, WHICH HIGHLIGHTS THE STRUCTURAL HEALTH BENEFITS OF TOUCHLESS INTERACTION. ULTIMATELY, IT PROVES THAT USERS HAVE A PRACTICAL, PHYSICAL INCENTIVE TO ADOPT GESTURE TECH, MOVING FAR BEYOND MERE CURIOSITY ABOUT A NOVEL GADGET

B. *Awareness and Prior Experience*

<div style="text-align: center;"> <p>Fig. 2. Participant's past experience with gesture control.</p> <img src="media/image2.png" alt="Bar chart of past experience" style="width: 4.79in; height: 2.76in;"> </div>

Awareness of gesture control was moderately widespread among participants: 44.4% reported having heard of the technology, 38.9% possessed a passing familiarity, and 16.7% had no prior knowledge whatsoever. However, actual usage lagged significantly behind awareness. Only 44.4% of the cohort had ever interacted with a gesture-based system, meaning more than half of those aware of the technology had never actually tried it. This discrepancy between awareness and practical adoption is a classic hallmark of emerging technologies. It aligns perfectly with the Unified Theory of Acceptance and Use of Technology (UTAUT) [23], which posits that cognitive awareness and behavioral facilitation are entirely distinct determinants of user adoption. Simply knowing a technology exists does not automatically translate into a willingness or opportunity to use it.

Among the 17 respondents who specified their information sources, YouTube emerged as the dominant channel (41.2%), followed by peer recommendations at 23.5%. Academic institutions and other social media platforms tied for a distant third at 11.8% each. This heavily indicates that public perception is shaped primarily by visual, consumer-oriented media rather than formal educational curricula. For developers and researchers aiming to boost public literacy around gesture control, short-form video content and digital demonstrations appear far more effective than traditional classroom instruction—at least within this specific demographic.

C. *Perceived Usefulness, Trust, and Innovation*

The scalar responses paint a picture of cautious optimism. When asked if gesture control could realistically replace a traditional mouse, 38.9% agreed, exactly half (50.0%) remained on the fence, and a tiny 11.1% disagreed. Crucially, though, nobody expressed strong opposition. We saw a very similar pattern regarding public utility, where 44.4% anticipated practical value, 38.9% were uncertain, and a combined 16.7% disagreed. This likely mirrors the intuitive grasp users have on the hygienic benefits of touchless interaction in public spaces, a theme well-documented in existing healthcare HCI literature [2]. Remarkably, two questions received zero negative responses. When evaluating whether gesture recognition feels innovative yet easy to learn, 44.4% agreed while 55.6% were unsure. Similarly, when asked if they would trust an AI-driven gesture system, 55.6% agreed and 44.4% remained neutral. The complete absence of outright rejection across both metrics reveals a baseline openness toward the technology, even before users gain hands-on experience. However, the lack of strong agreement on the trust query shows this openness is tentative. Trust here is highly conditional; users aren't inherently hostile toward AI gesture control, but they are waiting for the technology to prove its reliability.

Perceptions of utility and hardware readiness were equally balanced. Exactly half of the respondents believed that AI gesture recognition could streamline workflows and boost productivity, while the other half remained neutral—a finding that echoes research into gesture systems designed for assistive tech and motor-accessibility frameworks [4]. Regarding hardware, 44.4% agreed that a standard webcam is fully capable of driving an AI gesture interface. This is a fascinating data point because it closely aligns with contemporary engineering milestones [10], which prove that single-camera setups can substitute for expensive depth sensors. This alignment is critical: users aren't being asked to take a leap of faith regarding camera capabilities. Their intuitive assumptions are already matching up with what the engineering literature shows is possible.

D. *Adoption Barriers*

<div style="text-align: center;"> <p>Fig. 3. Worries people have when thinking about AI gesture control.</p> <img src="media/image3.png" alt="Bar chart of worries" style="width: 4.79in; height: 3.02in;"> </div>

Table III brings together the core anxieties and roadblocks highlighted by our participants. Honestly, both lists point to the exact same conclusion from slightly different angles. Performance is what matters most. System accuracy absolutely topped the charts—flagged as a major worry by 61.1% of users and a definitive barrier to adoption by another 50.0%. Sluggish response speeds and poor environmental lighting conditions followed closely behind, hovering as blockers for 33.3% to 50.0% of the group.

In stark contrast, the learning curve barely registered on their radar. It was the least mentioned issue across the entire questionnaire, sitting at a tiny 5.6%. Meanwhile, privacy concerns were right up there with technical performance. Half the cohort flagged data privacy as a general worry, and a full third viewed it as a hard barrier keeping them from using the tech.

Table 3 **CONCERNS AND BARRIERS TO ADOPTION**

Concer	**Concern %**	**Barrier**	**Barrier %**
Accuracy	61.1	Poor accuracy	50.0
Privacy	50.0	Slow response	50.0
Cost	38.9	Poor lighting	38.9
Camera quality	38.9	High CPU usage	33.3
Response speed	33.3	Privacy concerns	33.3
Learning curve	5.6	Prefer mouse	22.2

This alignment directly mirrors the technical hurdles highlighted in Section II's literature review. Interestingly, the baseline YOLO-CNN architecture notes that its classification accuracy slips to roughly 89% against cluttered backgrounds, documenting specific failures where simple hand waving is misread as pointing due to visual occlusions [10]. Our survey participants were never shown these engineering figures. Yet, they intuitively pinpointed tracking accuracy and ambient lighting as their primary anxieties. This overlap carries immense practical weight. It shows that ongoing engineering efforts to solve lab-based bottlenecks—like mitigating lighting shifts, handling obstructions, and slashing latency—happen to be the exact changes needed to win over real users. On the flip side, spending resources to make gesture controls easier to learn is a misallocation of effort since users don't view the learning curve as a major obstacle anyway.

Similarly, the dual presence of privacy as both a general worry and a hard barrier reflects broader social acceptability research [15], which reminds us that camera-driven interfaces are, by design, always watching. One respondent explicitly suggested that developers should "add privacy features like processing on the device." It is a sharp remark that mirrors a cutting-edge design trend. Edge computing—processing data locally on the machine rather than sending it up to the cloud—is precisely how modern gesture and wearable tech developers are trying to defuse user paranoia.

E. *Preferred Features*

Table IV maps out the specific mouse replacement features that participants actually want mapped to hand gestures. Page scrolling blew past everything else, coming out on top at a massive 83.3%. Drag-and-drop actions followed at a solid 61.1%. Interestingly, systemic adjustments—things like volume and brightness tweaks, taking screenshots, or executing a double-click—all clustered together in a tight bracket between 50% and 56%. Basic clicks, standard single-left clicks, and media playback buttons sat at the absolute bottom of the wish list. Even so, none of these lower-tier requests dropped below a third of the cohort, meaning there is still plenty of interest across the board for full-suite control.

Table 4 **PREFERRED GESTURE-MAPPED FEATURES**

Feature	**% Selecting**	**Gesture Type**
Scroll	83.3	Continuous / coarse
Drag and drop	61.1	Continuous / coarse
Volume control	55.6	Continuous / coarse
Screenshot	55.6	Discrete
Double click	50.0	Discrete / precise
Left click	44.4	Discrete / precise
Right click	38.9	Discrete / precise
Media control	33.3	Discrete

When analyzing the feature choices based on movement type, a pattern emerges where high-preference actions, such as scrolling and drag-and-drop, involve continuous, sweeping gestures, while lower-preference actions, like left and right clicking, require small, precise movements. This ranking isn't random; it directly aligns with the technical limitations in computer vision, where micro-gesture classification often fails, for example, confusing a "scissors" gesture with a pointing gesture [10]. Without technical knowledge, participants instinctively chose gestures that current systems handle most reliably, suggesting a strategy for Sections V and VI: early gesture interfaces should focus on large, continuous actions over precise, small-scale commands.

F. *Hardware Preferences and Adoption Intention*

When it comes to hardware preferences, the data shows a clear trend: people want to use what they already own rather than buying specialized sensors. Over half the group (55.6%) preferred routing gesture tracking through their smartphone camera, 38.9% favored a built-in laptop webcam, and a tiny 5.6% were willing to set up an external webcam. This strongly backs up the argument made against depth-sensor setups, which are highly reliable but require specialized hardware that the average user simply does not have [13]. For this cohort, everyday convenience and utilizing existing devices easily trumped the minor accuracy gains offered by high-end, dedicated peripherals.

<div style="text-align: center;"> <p>Fig. 4. How likely people said they would recommend and use gesture control.</p> <img src="media/image4.png" alt="Bar chart of adoption intention" style="width: 5.0in; height: 2.33in;"> </div>

The most striking signal in our entire dataset pops up right in the adoption-intention metrics. When asked if they would recommend gesture control to others, 55.6% said yes, while the remaining 44.4% gave a cautious maybe. Zero rejections. Then, we asked if they would actually use the technology if it were available right now; a resounding

66.7% said definitely, and the other 33.3% hovered in the unsure camp. Again, not a single person selected "definitely not." When you weigh these numbers against the technical anxieties mentioned earlier, it becomes incredibly clear that user hesitation isn't an ideological rejection of the tech. Rather, it is entirely conditional. The "maybe" and "not sure" groups shouldn't be written off as skeptics; they are simply an audience waiting for engineers to prove the system's accuracy and everyday reliability.

G. *Comparison with Previous Studies Focused on Perception*

Table V positions our current work against the handful of prior studies that prioritised user acceptance over raw algorithmic performance. Historically, most perception-focused research has either leaned on tiny sample sizes within tightly controlled laboratory walls [1] or bundled broad acceptability concepts into literature reviews without gathering fresh, empirical survey data [15]. This study occupies a unique, functional niche. It provides a compact but entirely original survey dataset that directly interrogates a highly specific use case—using vision-based hand gestures as a day-to-day replacement for the physical mouse—rather than debating the abstract merits of gesture interaction as a whole.

Table 5 **COMPARISON WITH PERCEPTION-FOCUSED STUDIES**

Study	**Primary Focus**	**Sample**	**Method**
Ahlström et al. [1]	Social acceptability, public gestures	Lab-based	Elicitation study
Koelle et al. [15]	Acceptability methods survey	Literature	Systematic review
Brunetti et al. [4]	Factory worker acceptance	Literature	Survey / review
This study	**Perception & adoption intent**	**n = 18, general users**	**Structured survey**

H. *Thematic Analysis of Open-Ended Responses*

Six distinct themes repeatedly surfaced when we parsed the qualitative answers regarding feature requests, necessary improvements, and the ultimate trajectory of human-computer interaction.

Accuracy was the most dominant theme by far. It popped up in almost every single response detailing what needs to be improved next. Participants explicitly demanded that developers "increase accuracy," "reduce errors," and "improve accuracy in low lights." This perfectly mirrors the quantitative rankings from Section IV-D. It adds real, empirical weight to the scalar data; users aren't just expressing a vague, abstract worry about AI, but rather a highly specific, clearly articulated engineering demand.

Voice control emerged as the most frequently suggested feature extension. Interestingly, several participants independently recommended merging gesture tracking with voice recognition instead of keeping them as isolated input methods. This implies that multimodal interaction—combining voice and gestures—is a direction that everyday users naturally expect, rather than an arbitrary concept forced by academic researchers. We revisit this path as a core future research vector in Section VIII.

Privacy and security were brought up both as standalone concerns and as part of a deeper anxiety regarding artificial intelligence. When we explicitly asked about fears surrounding AI, rather than gesture tech specifically, responses shifted toward data misuse, job displacement, and systemic AI abuse. There is a distinct line here. While gesture-specific anxieties are narrow and strictly technical (focusing on tracking precision and ambient illumination), broader AI anxieties are societal and ethical. Developers of gesture systems inevitably inherit this broader skepticism simply by association, even if their specific tracking algorithms pose no such threats.

Accessibility and natural interaction dominated the commentary regarding the future of human-computer interaction. Respondents painted a vivid picture of an ecosystem where interfaces become "more natural," "easier," and "user friendly." They envision a convergence of AI, voice, and AR/VR technologies working together to make computing "easier, faster, and more accessible for everyone." Meanwhile, **lighting issues** were singled out as a critical roadblock to fix immediately, with users demanding systems that "improve accuracy in lights." This direct feedback closely reflects the C2 varying-illumination stress tests documented in current computer vision literature [10].

Finally, the **real-world use cases** envisioned by the cohort were grounded entirely in daily convenience. Participants suggested deploying gesture control for delivering presentations, handling smart home appliances touch-free, navigating online classes, gaming, driving, and operating interfaces while cooking or managing dirty hands. These user-generated scenarios directly shape the real-world application frameworks we explore next in Section V.

I. *Synthesis*

When evaluated holistically, the survey data portrays a user base defined by cautious optimism, completely free of polarizing extremes. No single item on the entire questionnaire drew a majority negative response. More tellingly, the two central adoption queries elicited zero "no" responses. Yet, general consensus rarely pushed past standard agreement into outright enthusiasm. The specific pain points articulated by users—namely tracking accuracy, latency, ambient illumination, and data privacy—are functionally engineering-based, not ideological. This distinction is crucial. The primary bottleneck to widespread adoption is not a lack of user desire or an innate skepticism toward touchless tech.

Instead, users simply remain unconvinced that current systems are robust enough to handle daily workflows. This practical skepticism perfectly mirrors the ongoing focus on system reliability that drives the modern engineering literature outlined in Section II.

V. I. Synthesis

The real-world deployment channels explored below fuse established academic research with the explicit use cases volunteered by our participants in the open-ended text fields. Rather than an unvetted list of tech concepts, each domain is analyzed through its specific functional constraints.

A. Healthcare

Touchless interfaces are exceptionally critical inside operating theaters. In these high-stakes environments, strict sterility protocols mean surgeons cannot interact with traditional keyboards or monitors without contaminating their hands. Prior clinical trials tracking sterile gesture inputs confirm this utility [2]. The same logic extends cleanly to outpatient clinics, radiology labs, and intensive care units, where manipulating patient scans or navigating electronic charts without touching shared surfaces drastically curtails cross-infection risks.

B. Education

Two respondents independently flagged digital presentations and online learning modules as ideal proving grounds for gesture integration. In practice, a presenter can seamlessly advance slides, overlay digital ink annotations, or pause educational videos from across the room. They don't have to fiddle with laser clickers, stand stiffly by a podium, or break eye contact with their students. This fluidity aligns perfectly with the dynamic, spatial nature of modern interactive teaching methods.

C. Smart Homes

A participant explicitly requested gesture-driven controls for household lighting, media centers, and smart appliances. Within the smart home ecosystem, gesture recognition must actively compete with an already dominant, highly mature modality: voice assistance. Here, the distinct advantage of hand gestures lies in environments where vocal commands are counterproductive or entirely impractical. Think quiet nurseries at night, chaotic, noise-heavy living rooms, or any scenario where speaking aloud feels socially awkward or disruptive.

D. Gaming

Integrating hand gestures into gaming setups liberates players from the static constraints of gamepads and joysticks. It offers tactile, physical loops that heighten immersion in motion-based or virtual reality titles. This domain is also where foundational gesture lexicons—like pointing, waving, or rock-paper-scissors tracking—were first popularized. Our survey cohort actively highlighted this interactive layer as a preferred entertainment feature.

E. Manufacturing and Industry 4.0

Industrial factory workers frequently operate with greasy, gloved, or full hands, making traditional tactile inputs a logistical nightmare. Comprehensive reviews of worker-centric factory tech champion the pairing of mid-air gestures with large overhead displays [4]. However, field reports emphasize that assembly-line workers will only adopt these tools if they are entirely confident the system won't misfire during high-stakes, time-sensitive manufacturing tasks. This industrial reality maps perfectly onto our cohort's primary demand for flawless tracking accuracy.

F. Automotive

An automotive use case surfaced in the open-ended responses, but the existing literature couples this opportunity with a strict safety warning. Empirical driving studies reveal that complex mid-air gestures actually trigger higher cognitive loads and driver distraction than muscle-memory physical dials [11]. Consequently, in-vehicle gesture deployment should be strictly confined to low-effort, continuous adjustments—such as swiping to adjust climate control or twirling a finger to tweak audio volume—rather than complex menus that pull eyes off the road.

G. Public Kiosks and Shared Displays

A solid 44.4% of our sample viewed gesture controls as highly useful in public settings, driven entirely by the desire to avoid contaminated, shared surfaces. Digital ticketing terminals, mall information directories, and self-service checkout kiosks are textbook candidates for this tech. This is especially true in high-traffic transit hubs where hundreds of strangers touch the exact same screen interfaces back-to-back every single hour.

H. Accessibility

For individuals navigating severe motor impairments or restricted upper-limb mobility, vision-based hand tracking serves as a vital alternative to standard physical mice. Furthermore, long-standing research into automated sign-language translation proves how computer-vision frameworks can bridge critical communication gaps [6]. Half of our survey participants explicitly agreed that AI gesture tech is a powerful tool for accessibility and workplace equity. Because not a single respondent disagreed, this humanitarian application stands out as one of the most universally accepted paths for the technology.

VI. Challenges

The roadblocks facing gesture adoption stem from a mix of lab-proven bottlenecks and the raw anxieties voiced by our participants.

- **Accuracy: Precision remains the ultimate gatekeeper. It was flagged as a major worry by 61.1% of users and a hard barrier by 50.0% of the sample. The engineering literature confirms this anxiety; even highly optimized systems see their tracking efficiency dip to around 89% when forced to navigate cluttered backgrounds or visually ambiguous, overlapping hand postures [10]. Overcoming this hurdle requires shifting away from pristine laboratory tests toward aggressive, real-world deployment modeling.**
- **Lighting: Dimly lit rooms are kryptonite for computer vision. Both our survey data (where 38.9% cited poor lighting as a major hurdle) and foundational technical papers show that illumination vulnerabilities are a systemic issue, not an edge case [8]. Average homes, lecture halls, and offices rarely enjoy the immaculate, uniform lighting needed for flawless optical tracking.**
- **Privacy: Cameras are invasive by nature. Because vision-based systems must actively "see" their users, half of our cohort flagged privacy as a primary anxiety, with several explicitly demanding localized, on-device processing. Mitigating this fear isn't just about writing secure code; developers must actively communicate data-handling transparency to earn user trust.**
- **Cost: Perception doesn't always match reality. Interestingly, 38.9% of respondents worried about the financial cost of adoption, despite engineering breakthroughs proving that modern algorithms run beautifully on cheap, pre-existing consumer webcams [21]. This disconnect highlights a critical communication gap that researchers and marketers need to bridge.**
- **Response Speed: Sluggish tracking kills immersion. A third of the participants labeled latency as a dealbreaker. Because physical interactions require instantaneous visual loops, any noticeable input lag feels deeply unnatural. The engineering consensus establishes 25 frames per second as the absolute bare minimum for usable, real-time tracking [10].**
- **User Learning Curve: Teaching users new gestures is a non-issue. It was the least mentioned concern across the entire study, sitting at a minor 5.6%. This strongly suggests that development priorities should pivot away from oversimplifying gesture libraries and focus entirely on maximizing backend algorithmic stability.**
- **Camera Limitations: Basic webcams bring inherent hardware flaws, including narrow fields of view, poor dynamic range, and motion blur. Even though our participants overwhelmingly preferred using their existing, built-in hardware (Section IV-F), this convenience introduces severe optical bottlenecks that developers must learn to bypass via software optimization.**

VII. Limitations

We must interpret these findings within the structural boundaries of our study design. First and foremost, our sample size is tightly constrained to eighteen individuals (n = 18). Consequently, the data maps out preliminary behavioral trends rather than universally generalizable facts. Because we sampled from a single university community in Sri Lanka, the cohort lacks broad demographic diversity regarding occupational variance, age distribution, and baseline technical literacy.

Furthermore, our sample is heavily female-dominated and operates exclusively within the Windows ecosystem. This homogeneity is a direct artifact of convenience sampling rather than intentional research design, which naturally limits our insights regarding alternative operating systems. Finally, the study relies entirely on self-reported data. Most participants evaluated the technology based on conceptual imagination or brief historical exposure rather than long-term, daily interaction. Our qualitative data is similarly thin; the open-ended text fields drew significantly fewer responses (4 to 6 entries) than the scalar modules (17 to 18 entries), rendering our thematic synthesis exploratory rather than definitive.

VIII. Future Work

Our study's limitations outline a clear trajectory for future research. Expanding the participant pool to include a larger, globally distributed sample across multiple academic institutions and industry sectors would thoroughly stress-test our preliminary trends. A longitudinal study tracking user sentiment both before and after prolonged exposure to a live gesture framework would reveal whether adoption anxieties fade or intensify over time.

Additionally, blending quantitative surveys with empirical usability testing—where users complete standardized computing tasks using gesture controls—would validate whether self-reported accuracy concerns match actual user error rates. Given the strong qualitative demand for voice integration, developing and testing a multimodal

voice-gesture pipeline stands out as an incredibly promising design vector. Exploring hand tracking within augmented and virtual reality (AR/VR) spaces, where spatial interactions are already native, would also yield valuable UX insights. Lastly, conducting targeted user studies with motor-impaired individuals is essential to empirically substantiate our accessibility claims.

IX. Conclusion

This paper tackles a fundamental question that traditional, performance-heavy computer vision literature routinely ignores: do the people these systems are built for actually want to use them? Our survey of a university community in Sri Lanka revealed a population with low practical exposure to touchless interfaces; less than half had ever used a gesture system. Yet, they are remarkably receptive. Faced with high rates of physical discomfort from traditional keyboards and mice, two-thirds of the cohort expressed a clear willingness to adopt gesture control, and not a single respondent rejected the concept outright.

The remaining barriers to adoption are strictly technical, not attitudinal. Across every metric, tracking accuracy, response latency, environmental lighting resilience, and data privacy consistently outranked the learning curve as primary user anxieties. Fortuitously, these exact performance bottlenecks are the very issues currently dominating the engineering research pipeline. This alignment provides a clear mandate for the HCI community: the most direct path to closing the user adoption gap is to keep closing the algorithmic reliability gap in the lab. Rather than redesigning gesture vocabularies or over-investing in basic tutorials, developers must focus on technical stability. Human perception and engineering precision must advance hand-in-hand before gesture recognition can transition from a novel boardroom gimmick into a trusted, daily productivity tool.

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